UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

UCCD2303 DATABASE TECHNOLOGY
(EVENING AND WEEKEND CLASSES)

SATURDAY, 5 OCTOBER 2024 TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
DIGITAL ECONOMY TECHNOLOGY

Instruction to Candidates:

This question paper consists of **Section A** and **Section B.**

Section A consists of **THREE (3)** questions. Candidates are required to **answer ALL** the questions.

Section B consists of **TWO (2)** questions. Candidates are required to **answer only ONE (1)** question. Should a candidate answer more than **ONE (1)** question in **Section B**, marks will only be awarded for the **FIRST** questions in the order the candidate submits the answers.

Candidates are not allowed to use a calculator.

Answer questions only in the answer booklet provided.

Section A: Answer ALL QUESTIONS. [Total: 75 Marks]

Q1. (a) Elaborate the Three-level ANSI-SPARC Architecture with a diagram. (6 marks)

(b) State **THREE (3)** differences between a BASE RELATION and a VIEW. (6 marks)

(c) Explain how inheritance enables a SUBTYPE entity to inherit the attributes and relationships of the SUPERTYPE entity by using **ONE (1)** example. (6 marks)

(d) Consider the following statements:
"A car is made up of an engine. However, an engine can exist on its own without a car. An engine of a car can be moved to another car."

Determine whether GENERALIZATION or AGGREGATION or COMPOSITION exists in the given scenario above. Provide **FOUR (4)** justifications for your answer. (7 marks)

[Total: 25 marks]

Q2. (a) Justify how consistency rule can be enforced in a database between a Customer table and an Order table. (4 marks)

(b) Determine whether deadlock exists from transactions T1 to T5 as shown in Figure 2.1 for item A, B, C and D with explanations

T1	W(A)									R(D)
T2		R(A)							W(A)	
T3			W(B)					R(C)		
T4				R(C)			W(A)			
T5					R(D)	W(B)				

Figure 2.1 Operations of transaction T1 to T5

W denotes Write
R denotes Read

(7 marks)

(c) Explain the **FOUR (4)** basic properties in the context of database transactions to ensure all the operations of a database transaction are reliable and consistent. Illustrate each property with **ONE (1)** example. (8 marks)

(d) A database can become useless due to hardware or software failure and the different failure situations may require different recovery techniques. As an IT consultant, you are required to explain the differences of the **THREE (3)** main recovery techniques of a database. (6 marks)

[Total: 25 marks]

Q3. Answer all the questions below based on the given scenario in Figure 3.1.

Given scenario:

> As a database designer, you are required to develop a database system that models soccer teams, the games they play, and the players in each team.
>
> In the design, the following information are required to be captured:
> A set of teams, each team has an ID (unique identifier), name, main stadium, and to which city this team belongs.
> Each team has many players, and each player belongs to one team. Each player has a number (unique identifier), name, date of birth, start year, and shirt number that he uses. Teams play matches, in each match there is a host team and a guest team. The match takes place in the stadium of the host team.
>
> For each matches, the data that needs to be tracked includes the date on which the game is played, the final result of the match, the players participating in the match. For each player, how many goals he scored, whether or not he took a yellow card, and whether or not he took a red card.
>
> During the match, one player may substitute another player. The model needs to capture this substitution and the time at which it took place. Each match has exactly three referees. For each referee, the data to be captured are the ID (unique identifier), name, date of birth and year of experience. One referee is the main referee and the other two are assistant referees.

Figure 3.1 A scenario about a soccer match

(a) Identify the FUNCTIONAL DEPENDENCY. (5 marks)

(b) Identify the CARDINALITY. (5 marks)

(c) Construct **ONE (1)** Entity Relationship Diagram (ERD) with attributes to capture the requirements for the scenario. State any assumptions you have that affect your design.
Hints: Use a surrogate key and Crow's Foot Notation. (10 marks)

(d) Write the Third Normal Form (3NF). State the primary key(s) and foreign key(s) for each relation in your answers. (5 marks)

[Total: 25 marks]

Section B Answer ONLY ONE (1) question. [Total: 25 marks]

Q4. (a) Answer ALL the questions below based on the database scheme in Figure 4.1:

Customer (cid, name, city) Product (pid, name, price) Purchase (pid, cid, store)

Figure 4.1 A sales scheme

(i) Draw **ONE (1)** Relational Algebra Tree for the following SQL query. (6 marks)

```
SELECT DISTINCT x.store
FROM Customer y, Purchase x
WHERE x.cid = y.cid
AND y.city = 'Seattle';
```

(ii) Draw **ONE (1)** Relational Algebra Tree for the following SQL query. (9 marks)

```
SELECT z.city, sum(x.price)
FROM Customer z,, Product x, Purchase y
WHERE x.pid = y.pid
AND y.cid = z.cid
AND y.store = 'Wal-Mart'
GROUP BY z.city
HAVING count(*) > 100;
```

(b) Explain the **FOUR (4)** major components in the Reference Architecture of a Distributed Database Management Systems (DDBMS). (4 marks)

(c) There are several advantages for dividing the data into fragments to be stored in different computers. However, there are also some issues related to the placement of data in a distributed database. Provide **ANY THREE (3)** primary drawbacks of having data fragmentation in a distributed database (6 marks)

[Total: 25 marks]

Q5. (a) Explain how commodity servers are related to Hadoop. (6 marks)

(b) Even though Big Data has brought numerous benefits across industries and society, it also presents some challenges. Predict **ANY** possible **THREE (3)** challenges in the use of Big Data. Justify your answers. (6 marks)

(c) Both on-premises database and cloud database can provide your business with the IT infrastructure it needs. The model to be used likely depend on the level of security needs in order to meet compliance standards as well as depend on the cost structure of choice. As the IT consultant, you are required to explain the following questions raised by your clients.

(i) State the definition of Cloud Database. (2 marks)

(ii) Provide **ANY FIVE (5)** advantages of a cloud database over a local on-premises database with justifications. (5 marks)

(d) "Not only SQL" (NoSQL) is a non-schema alternative to Structured Query Language (SQL) to store, process, and analyse extremely large amounts of unstructured data. How does NoSQL differ from SQL in data processing? Justify your answers. (6 marks)

[Total: 25 marks]